

How do Archaeologists Identify Domestication?



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Domestication is fundamentally a genetic, evolutionary process that results in physical, or phenotypic, changes in behavior (in animals) or anatomy (in plants and animals). As a result, DNA analysis—especially the analysis of ancient DNA from archaeological specimens—is best suited to identifying the underlying genetic changes in plants and animals. Ancient-DNA analysis is, however, very expensive and only works on particularly well-preserved bones and seeds. Archaeologists have a variety of methods for identifying the morphological changes that accompany domestication, but those vary between the study of animals and plants.

Markers of Domestication in Animals

A challenge in identifying the domestication of animals from archaeological remains is that only animal bones are usually preserved, but as you learned earlier in Prof. Schaal's lecture, the most noticeable changes in animal morphology that result from domestication are to soft tissues, such as cartilage and musculature. Since the primary selection force was for behavioral change, durable skeletal markers of domestication are few. One way we can use bones to see a change in human relationships with animals is to identify the age and sex structure of animals killed and eaten by people. In a domesticated population, males are often killed much younger than females, since they do not produce milk, and only a few are needed to reproduce.

But in some cases, domestication only goes so far to change a species. Watch a short TED-Ed video on the domestication history of cats to see some of the alternative patterns of behavioral and genetic changes that accompanied a species domesticated on a different trajectory from that of dogs. As you watch, compare the information with the notes you took earlier on the “pathways” to animal domestication in Prof. Schaal's talk. On which pathway were cats domesticated? Does that explain some of their idiosyncratic behaviors today?



TED-Ed



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Markers of Domestication in Plants

Different plants respond to long-term human intervention in their reproductive cycles and ecology in different ways. This variety is dependent on both the underlying genetic structure and physical form of the plants, and the ways in which humans disrupted their natural ecology and reproductive cycles.

One of the leading scholars working to understand how plants have adapted to domestication by humans is Prof. Dorian Fuller of University College London. Watch his lecture embedded below (about 35 minutes long, ends at 39:00 in the video), to learn more about the ways that plants have changed during domestication in three separate regions: China, the Fertile Crescent of southwestern Asia, and Sub-Saharan Africa. As you watch the video, consider these questions:

1. What are the common responses of plants in these three scenarios?
2. What was the cause of domestication? Did people “decide” to domesticate plants?
3. How fast or slow was the process of plant domestication?

